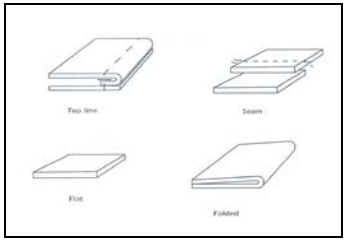
STANDARD OPERATING PROCEDURE OF Resistance to Stretching and Flexing VF-D06

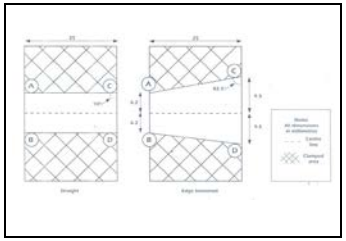
Lab /SOP/ 22



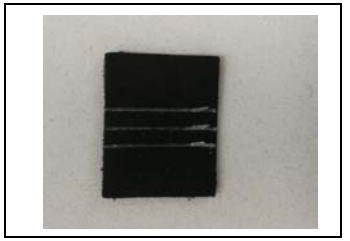
Upper Material Stretch and Flex Machine
Optical magnifier or stereoscopic microscope



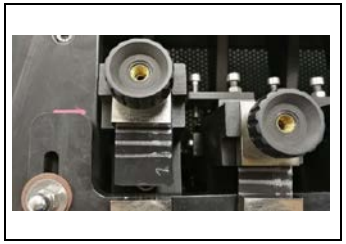
Prepare the test sample, either cut from completed item of footwear or produced specifically for testing.



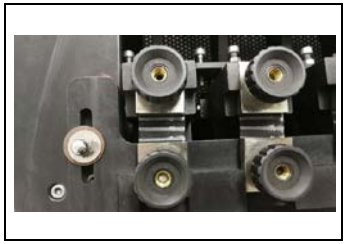
Sample size 38±2mmx25±2mm
Draw a line across the width of the test specimen, through its center and parallel to its shorter edges.
Draw a further two lines parallel to the center line, one 6.2 ±0.2 mm from either side of the center line.
Mark one of these lines AC and the other BD.



From footwear uppers: test three (3) samples
Sheet material: test three (3) samples for each direction.



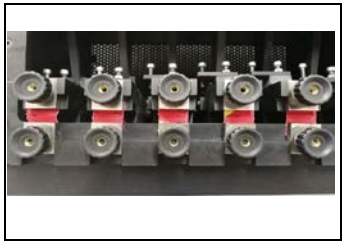
Loosen the stationary clamp and the moveable clamp.
Adjust the machine so that the clamps are as far as possible.
Adjust the machine so that the clamps are as far as possible.
Place the end of the test specimen closest to the line AC into a moveable clamp and insert



Fully tighten the moveable clamp, while taking care not to overtighten it and damage the test specimen.
Adjust the machine so that the clamps move together.
Stop when the second line, marked BD, on the test is coincident with the edge of the stationary clamp.



Reset all the counters to zero, and set the desired number of cycles.
Inspect the samples after every 5000 cycles during the first 20,000 cycles, after 20,000 cycles inspect the samples every 10,000 cycles.



If the damage of the specimen is considered to be unacceptable, then remove the test specimen from the flexing machine and record the number of flexing cycles to which the specimen failed.
Appearance of any crease, cracking, splitting or tearing which has occurred.
Report:Number of flexing cycles.

Prepared by	Approved by	REVISION		DATE	